Blockchain Applications: Practice Exam

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Instructions. This practice exam covers Blockchain Analytics (Lectures 1–8): 38 multiple-choice questions, each with four options and one correct answer, one point each, no negative marking. Questions 1–23 test concepts and the reading of course output; questions 24–38 are built on three short cases, each followed by five questions. No calculation beyond one-step mental arithmetic is needed. The real exam allows no books, notes or other materials, lasts 60 minutes and has 30 questions: 15 on Crypto Economics and 15 on Blockchain Analytics, including one case with 5 questions.

All numbers come from the lectures (EOD Historical Data, EODHD, prices to 23 October 2026; Lecture 3 to 30 October 2026). Abbreviations: VaR Value-at-Risk (reported as a positive loss, VaR $= -q$), ES expected shortfall, GARCH generalised autoregressive conditional heteroskedasticity, IGARCH integrated GARCH, HS historical simulation, EWMA exponentially weighted moving average, bp basis point (0.01%), HHI Herfindahl–Hirschman index, AUC area under the receiver operating characteristic curve, CAR cumulative abnormal return, DSR deflated Sharpe ratio, QS quantile score, DM Diebold–Mariano, CRIX CRypto IndeX, MV market value.

1. The estimate $\hat{H} = D \cdot 2^{32}/600$, with D the Bitcoin mining difficulty, measures (a) transactions per second; (b) hashes computed per second to secure the network; (c) the number of active miners; (d) the growth rate of the money supply.

2. The Herfindahl–Hirschman index of the five largest coins is 0.71. This means (a) 71% of coins are stablecoins; (b) the market is close to perfect competition; (c) market value is highly concentrated in a few coins; (d) the five coins have correlation 0.71.

3. After the 2024 halving, Bitcoin issuance inflation is about 0.82% a year. This follows from (a) a fixed supply rule: the block reward halves every 210 000 blocks, towards 21 million BTC; (b) a decision of the miners each year; (c) the hash rate; (d) the price of Bitcoin.

4. The Jarque–Bera statistic for daily Bitcoin returns is 29 620. Therefore (a) returns are normally distributed; (b) the variance is infinite; (c) returns are independent; (d) normality is rejected, and a VaR based on the Normal distribution understates crash risk.

5. Ljung–Box $Q(10)$ is 212.7 for squared returns r_t^2 and 20.2 for returns r_t. This shows (a) returns can be predicted profitably; (b) volatility clustering is much stronger than any autocorrelation of returns; (c) returns are independent; (d) returns are normal.

6. Robust variance-ratio tests do not reject weak-form efficiency in any sub-period. The correct conclusion is (a) prices are proven to follow a random walk; (b) there are no bubbles; (c) the data give no evidence of return predictability at the tested horizons; (d) returns are normal.

7. A log-periodic power law fits the 2017 Bitcoin bubble with $R^2 \approx 0.98$. Hence (a) a good in-sample fit does not give a reliable forecast of the crash date; (b) the crash date is known precisely; (c) the model is rejected; (d) the bubble did not exist.

8. On the reselection day of 30 September 2026 the divisor of the CRIX replica is recomputed from 214.67 to 214.83 million, while the index stays at 9 521.15. The divisor is adjusted (a) to convert coin prices into US dollars; (b) to cap the weight of Bitcoin below 50%; (c) so that a change of the basket does not make the index jump: only price changes move it; (d) to remove the past returns of coins that failed.

9. Five coins bought with equal weights on 1 January 2018 earn a compound annual growth rate (CAGR) of 17.6% to October 2026 if they are the five largest coins of 2026, but 7.1% if they are the five largest coins of 1 January 2018. The gap shows (a) that small coins always beat large coins; (b) that equal weights are better than market-value weights; (c) that the 2018 data are wrong; (d) survivorship bias: choosing coins with information from 2026 overstates what an investor could have earned in 2018.

10. The spot Bitcoin fund IBIT has a tracking error (TE) of 3.5% a year against Bitcoin priced at 16:00 New York time, but 20.6% against Bitcoin's close at midnight, Coordinated Universal Time (UTC). This shows (a) the fund holds futures instead of Bitcoin; (b) TE must be measured against the benchmark at the time the fund is priced; the fund itself did not change; (c) the fund's fee is 20.6% a year; (d) the fund does not follow Bitcoin.

11. For Bitcoin, GARCH(1,1)-t gives $\hat{\alpha} + \hat{\beta} = 1.000$. This means (a) volatility does not cluster; (b) volatility shocks do not die out in this sample (integrated GARCH); (c) a volatility shock halves in 36 days; (d) returns are independent.

12. A 1% VaR model has 45 breaches where 24.5 are expected; the Kupiec test has $p < 0.001$. The model (a) is too conservative; (b) passes the backtest; (c) underestimates risk; (d) has dependent breaches.

13. With Normal-based expected shortfall, realised Bitcoin losses beyond VaR are 26–31% larger than predicted. This means (a) the VaR level is wrong; (b) ES is always larger than VaR; (c) the Normal model is too conservative; (d) the Normal distribution understates how deep losses go once the VaR is breached.

14. During the SVB weekend USDC fell to −1 036 bp, while the value of a token with zero recovery on the SVB deposit was −825 bp. The extra discount reflects (a) a liquidity premium: redemption was impossible while banks were closed; (b) the expected credit loss; (c) a data error; (d) a change of the peg.

15. USDC has the smallest 1% quantile of USDT, USDC and DAI (−7.6 bp) but the largest kurtosis (1 045). Together these numbers say (a) USDC is the safest coin in every respect; (b) USDC is usually very close to the peg, but its risk is concentrated in rare, extreme episodes; (c) the data contain an error; (d) USDC deviations are normally distributed.

16. A Curve StableSwap pool with amplification $A = 2000$ quotes USDC at 0.997 while 90% of the pool is USDC. This indicates (a) USDC is safe, since the price is close to \$1; (b) a balanced pool; (c) the pool is becoming one-sided, a sign of stress the price does not yet show; (d) a data error.

17. A price-only alarm for stablecoin depegs has AUC 0.90 on test hours but detects none of the 8 test events in time. The explanation is (a) AUC is always misleading; (b) the data are wrong; (c) 8 events are too many; (d) a high AUC comes from hours inside episodes, while an alarm must fire before an episode starts.

18. In the stablecoin depeg pressure index (DPI) a higher average selected Quantile Lasso penalty $\hat{\lambda}$ indicates (a) stronger co-movement in the lower tail, with fewer but stronger links; (b) more active predictors; (c) lower volatility; (d) a better forecast.

19. The Quantile Lasso equations behind the DPI fit the lower tail of stablecoin deviations well in sample, yet held-out forecast skill is negative. The lesson is (a) the model is wrong; (b) in-sample fit is not out-of-sample forecasting ability; (c) more data always help; (d) the tail is normal.

20. In a Diebold–Yilmaz spillover table, $\mathrm{NET}_{\mathrm{DOGE}} = -18.6$. This means (a) Dogecoin fell 18.6%;

(b) Dogecoin causes the other coins; (c) Dogecoin receives more forecast-error variance from the others than it transmits; (d) Dogecoin is uncorrelated with the market.

21. The Merge has CAR$[-1, 1] = -11.1\%$ ($t = -3.09$); across 13 Ethereum upgrades the mean CAR is -0.18% ($p = 0.92$). The correct conclusion is (a) single upgrades can move the price strongly, but there is no common revaluation; (b) upgrades always destroy value; (c) the Merge result is an error; (d) all upgrades are significant.

22. XGBoost classifies tomorrow's Bitcoin direction with 78.3% accuracy in sample and 51.3% out of sample; 50.6% of days are up days. This is (a) strong evidence of predictability; (b) overfitting: the model learnt noise; (c) a data error; (d) proof of market inefficiency.

23. The best of 810 trading rules has Sharpe ratio 0.84 in 2018–2022; its deflated Sharpe ratio is 0.72. Hence (a) the rule is significant at 1%; (b) the rule will earn 0.84 in the future; (c) the Sharpe ratio is wrong; (d) once the number of trials is counted, the rule is not significant.

Case 1: Bitcoin Volatility and VaR Backtest

GARCH(1,1)-t	$\hat{\omega}$	$\hat{\alpha}$	$\hat{\beta}$	$\hat{\alpha}+\hat{\beta}$	half-life	$\hat{\nu}$
Bitcoin	0.144 (0.058)	0.076 (0.012)	0.924 (0.012)	1.000	∞	3.13 (0.15)
S&P 500	0.035 (0.009)	0.162 (0.022)	0.819 (0.021)	0.981	36	5.85 (0.71)

1-day VaR 1%, Bitcoin, 2020–2026	hits	rate (%)	Kupiec p	indep. p	last 250	Basel zone
Normal, 250 days	45	1.83	< 0.001	0.850	5	amber
HS, 500 days	25	1.02	0.924	0.256	3	green
EWMA, Normal	51	2.08	< 0.001	0.952	5	amber
GARCH(1,1)-t	28	1.14	0.491	0.329	2	green

Daily log returns in %; robust standard errors in brackets; 2 453 test days, 24.5 hits expected. Mean VaR: HS 8.64%, GARCH-t 8.12%. Kupiec p: test of a 1% breach rate; indep. p: test that breaches do not cluster.

24. The degrees of freedom $\hat{\nu}$ are 3.13 for Bitcoin and 5.85 for the S&P 500. This shows that (a) Bitcoin volatility is less persistent than S&P 500 volatility; (b) Bitcoin returns have about half the variance of S&P 500 returns; (c) standardised Bitcoin shocks stay heavy-tailed after GARCH filtering; (d) GARCH filtering removes the heavy tails of both return series completely.

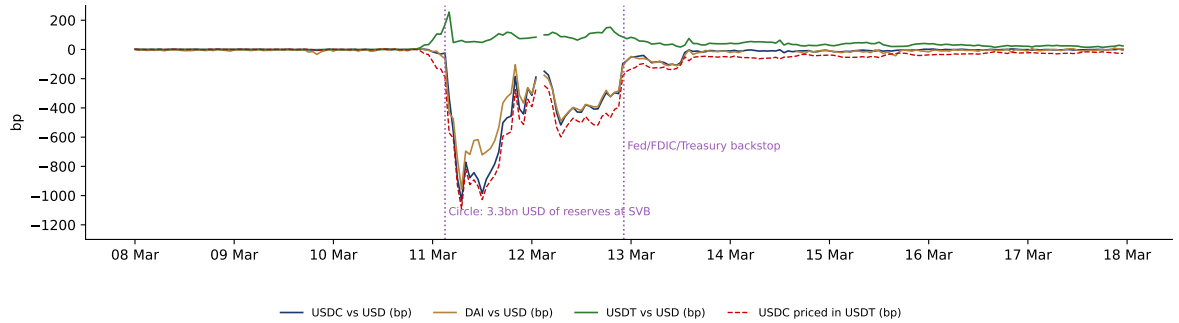
25. Which caveat applies to the Bitcoin persistence estimate $\hat{\alpha}+\hat{\beta}=1.000$? (a) it can also reflect structural breaks between very different market phases; (b) it proves that Bitcoin volatility never returns to a long-run mean; (c) it is invalid because robust standard errors were used in estimation; (d) it implies a half-life of 36 trading days, exactly as for the S&P 500 index.

26. The Normal and EWMA-Normal VaR have Kupiec $p < 0.001$ but independence p of 0.850 and 0.952. The justified reading is (a) their breaches cluster in time, so the volatility filter reacts too slowly; (b) they hold too much capital, since the VaR is breached far too rarely in the sample; (c) they pass the backtest, since the independence test is not rejected; (d) they breach too often but not in clusters: the Normal quantile is too thin.

27. A risk manager must choose between HS and GARCH-t, both in the green zone. Which argument favours GARCH-t? (a) its breach rate of 1.14% is closer to the promised 1% than that of HS; (b) it gives correct coverage with a lower mean VaR (8.12% vs 8.64%); (c) it has more breaches in the last 250 days than HS; (d) it is the only model whose Kupiec test is not rejected.

28. Which further evidence would best separate HS from GARCH-t? (a) a longer list of breach rates computed on exactly the same test days; (b) a Jarque–Bera test of normality on the raw Bitcoin returns; (c) quantile scores with a DM test, plus ES backtests by sub-period; (d) the number of breaches in the last 250 days of the sample.

Case 2: A Stablecoin under Stress



Hourly deviations from \$1 in bp around the Silicon Valley Bank (SVB) failure, March 2023. Circle held 3.3 bn USD of about 40 bn USD reserves at SVB (8.25%). Minimum: USDC −1 036 bp and DAI −955 bp at 07:00 UTC on Saturday 11 March. Early-warning model of the lecture (paper): 72% of test events detected, on average 14 hours ahead, with 2.5 false alerts a month; price-only alarms meet the false-alert budget only by detecting almost no events.

29. If Circle had lost its entire SVB deposit (8.25% of reserves), the fair value of one USDC would deviate from \$1 by (a) −330 bp; (b) −825 bp; (c) −1 036 bp; (d) −211 bp.

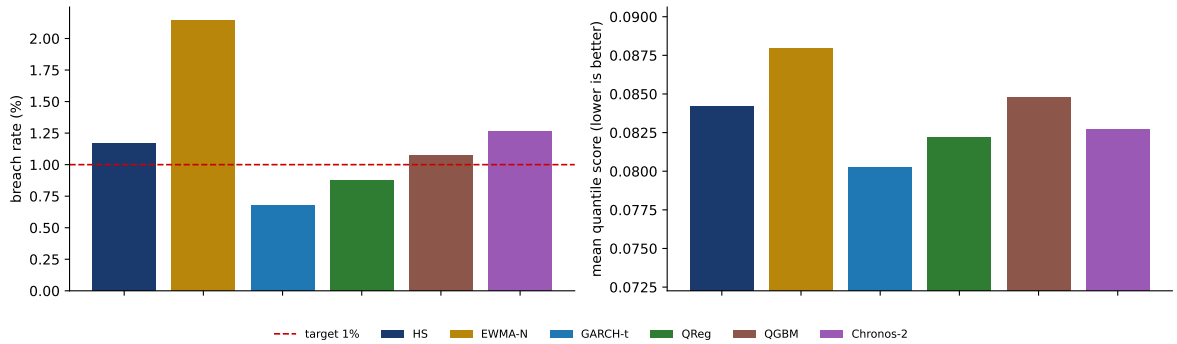
30. USDC traded down to −1 036 bp while about 92% of its reserves were untouched. The episode shows that (a) the remaining reserves were too small to back the token; (b) the market priced exactly the expected credit loss on SVB; (c) a fully backed stablecoin cannot deviate from \$1 for more than a few hours; (d) a peg can fail when redemption is closed, even with most backing intact.

31. DAI fell to −955 bp at the same hour, while USDT rose above \$1. The best explanation is (a) much of DAI's collateral was USDC, and holders fled to the other dollar coin; (b) DAI held its own deposits at SVB, while USDT was backed by the Federal Reserve; (c) DAI and USDT are both issued by Circle, so they moved in opposite ways; (d) the DAI move is a data error, since a decentralised coin cannot depeg.

32. The early-warning model detects 72% of test events, on average 14 hours ahead, with 2.5 false alerts a month. The correct reading is (a) 72% of all test hours were classified correctly by the model; (b) each alert is followed by a depeg within the next 14 hours; (c) 72% of test episodes were flagged before they began; (d) the model will detect exactly 72% of future depeg episodes.

33. Why monitor the composition of a StableSwap pool and not only its price, and what is the limitation? (a) the price moves first, but composition is cheaper to observe on chain; (b) composition moves first, but it also shifts for benign reasons (incentives); (c) composition and price move together, so either one is sufficient; (d) composition moves first, and any shift in it signals a coming depeg with certainty.

Case 3: Machine Learning and Risk Forecasts



model	breach %	Kupiec p	QS	DM p*
HS	1.17	0.60	0.0842	0.45
EWMA-N	2.14	0.001	0.0880	0.18
GARCH-t	0.68	0.28	0.0803	–
QGBM (boosted trees)	1.07	0.82	0.0848	0.36
Chronos-2 (pretrained)	1.27	0.41	0.0827	0.54

1-day VaR 1% for Bitcoin, January 2024 – October 2026: 1 027 test days, 10.3 breaches expected. *DM p: test of equal mean QS against GARCH-t. QS: mean quantile score, lower is better. QGBM: quantile gradient boosting machine. Chronos-2 was released in October 2025.

34. GARCH-t has the lowest QS (0.0803) although only 0.68% of days are breaches. This is because (a) a lower QS means more breaches, so GARCH-t is in fact too aggressive; (b) the QS rewards the model whose breach rate is closest to 1%; (c) GARCH-t fails the Kupiec test, so its QS cannot be used; (d) the QS judges daily closeness to the quantile, not the breach count.

35. All DM p-values against GARCH-t are 0.18 or higher. The justified conclusion is (a) we do not reject equal mean QS, so the ranking is not conclusive; (b) GARCH-t is significantly better than every other model; (c) all five models give the same forecast on every day; (d) HS is significantly worse than GARCH-t at the 45% level.

36. EWMA-N fails the coverage test (Kupiec $p = 0.001$), yet its QS is not significantly worse than that of GARCH-t (DM $p = 0.18$). Both can hold because (a) the Kupiec test is invalid for models with Normal quantiles; (b) Kupiec counts breaches, while the DM test on QS has little power here; (c) the DM test checks breach clustering, not the breach count; (d) EWMA-N has too few breaches, which the quantile score does not penalise.

37. Should EWMA-N be used for a 1% VaR? (a) yes, since its QS is not significantly worse than that of the GARCH-t model; (b) no: its volatility filter should be replaced by a constant variance; (c) no: failed coverage is a problem in itself, and the fix is the distribution; (d) yes, as long as the test period is extended to more than 2 000 days.

38. Chronos-2 performs as well as GARCH-t without any fitting to Bitcoin. The main concern about this comparison is (a) its QS of 0.0827 is significantly worse than that of GARCH-t; (b) a pretrained model cannot produce quantile forecasts at all; (c) it was not fitted to Bitcoin, so it cannot contain any look-ahead bias; (d) its training data may include Bitcoin prices from the test period.